

Urban Heat Islands

Student sheet

Name: _____

Date: _____

1. What I see on the screen

The screen shows **Temperature** (°C) and **Humidity** (%), with the temperature chart below them. You start with **Start**; the green button, bottom right, moves you on, **Back** takes you back a step, and the last green button reads **Finish**. The on-screen table **does not save: this sheet is the record**.

I measure the air above the surface, not the surface. Asphalt can be 60 °C, but the air above it, much less.

The protocol — five rules, at every location: (1) a height of ____ cm, the same everywhere, measured; (2) a white screen above the sensor, *including* in the shade; (3) **a full two minutes**, timed; (4) everyone at **two metres** while it settles; (5) **the time noted** with the number.

2. What I do on my own

Weather: _____

Start: _____

End: _____

First I predict. I do not change the prediction after I measure. 1. The warmest location will be _____, the coolest _____. 2. The difference between them will be _____ °C. 3. Indoors will be ☐ warmer ☐ cooler than outdoors, because _____

Now I measure. I wait two minutes at each location. **Row 7 is compulsory:** it is the same location as row 1, measured at the end.

#	Location (surface, sun or shade)	Time	T (°C)	RH (%)
1	Grass in shade — reference			
2	Asphalt in sun			
3	Indoors (classroom or corridor)			
4				
5				
6				
7	Grass in shade — return			

The map. Write the *corrected* temperature in each location's box, then colour blue for the coolest and red for the warmest. The schoolyard plan, if you received one, is glued to the back of the sheet.

THE BUILDING (indoors) no. ____ $T = \text{_____}^{\circ}\text{C}$	ASPHALT IN SUN no. ____ $T = \text{_____}^{\circ}\text{C}$	ASPHALT IN SHADE no. ____ $T = \text{_____}^{\circ}\text{C}$
GRASS IN SHADE (ref.) no. ____ $T = \text{_____}^{\circ}\text{C}$	GRASS IN SUN no. ____ $T = \text{_____}^{\circ}\text{C}$	BESIDE THE SUNLIT WALL no. ____ $T = \text{_____}^{\circ}\text{C}$

$$\Delta T \text{ (grass in shade} \rightarrow \text{asphalt in sun)} = \underline{\hspace{1cm}} - \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \text{ } ^\circ\text{C}$$
 Warmest:
$$\Delta T \text{ (indoors} \rightarrow \text{outdoors)} = \underline{\hspace{1cm}} - \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \text{ } ^\circ\text{C}$$

Coolest:

My drift: _____ °C. Biggest difference between locations: _____ °C. So the difference $\square$ is detectable $\square$ is not detectable with our setup. “*Not detectable*” does not mean “does not exist”.

I predicted that, and the data show that _____

The coolest location was the coolest because _____

I propose that our schoolyard should _____

Exit ticket

One thing that surprised you today, and one thing you would do differently if you repeated the measurement.